

Some Details

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1 Normalization of Polarization Vectors

First, we give the definition of polarization vectors which is written in terms of adjoint representation

$$\varepsilon_a^\mu = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p^J \rangle (\sigma_a)^I_J}{4(p \cdot r)} \quad (1)$$

here a runs from 1 to 3. We can use antisymmetric tensor to rewrite this equation, like

$$\varepsilon_a^\mu = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_K \rangle \epsilon^{JK} (\sigma_a)^I_J}{4(p \cdot r)} \quad (2)$$

here we choose the left-multiplication convention.

Then, a 2×2 symmetric matrix can be defined like

$$(\sigma_a)^{IJ} = \epsilon^{JK} (\sigma_a)^I_K. \quad (3)$$

This means that the original polarization vector can be understood as

$$\begin{aligned} \varepsilon_a^\mu &= -\frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_J \rangle}{4(p \cdot r)} (\sigma_a)^{IJ} \\ &= -\varepsilon_{IJ}^\mu (\sigma_a)^{IJ} \end{aligned} \quad (4)$$

here I, J are fundamental little group indices, runs from 1 to 2.

And we know that the reasonable polarization vector for physical state should be symmetric with little group indices, it means that we need to symmetrize the indices I and J as $(IJ) = \frac{1}{2}(IJ + JI)$. Because of the symmetry of $(\sigma_a)^{IJ}$, so it can be obtained that

$$\varepsilon_a^\mu = -(\varepsilon_{(IJ)}^\mu + \varepsilon_{[IJ]}^\mu) (\sigma_a)^{IJ} = -\varepsilon_{(IJ)}^\mu (\sigma_a)^{IJ} \quad (5)$$

which means we only need to consider the symmetric part of ε_{IJ}^μ .

Now, we want to compute the normalization for ε_{IJ}^μ , and after multiplying the symmetric matrix, it should go back to the normalization

$$\varepsilon_a^\mu \varepsilon_{b,\mu} = -\delta_{ab} \quad (6)$$

The method to compute the normalization is just multiplying two polarization vectors directly, like

$$\varepsilon_{(IJ)\varepsilon_{(KL),\mu}}^\mu = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_J \rangle}{4(p \cdot r)} \times \frac{\langle p_K | (r \cdot \Gamma) \Gamma_\mu | p_L \rangle}{4(p \cdot r)} \quad (7)$$

here we need to use a special 5d equality

$$(\Gamma^\mu)_A^B (\Gamma_\mu)_C^D = -2\Omega_{AC}\Omega^{BD} + 2\delta_A^D \delta_C^B - \delta_A^B \delta_C^D \quad (8)$$

and the dirac operator can be decomposed to

$$\not{p} = p \cdot \Gamma = p_A^B = |p_I\rangle_A \langle p^I|^B \quad (9)$$

which help us to rewrite the equation (7) to be

$$\begin{aligned} \varepsilon_{IJ\varepsilon_{KL,\mu}}^\mu &= \frac{\langle p_I |^A | r_M \rangle_A \langle r^M |^B (\Gamma^\mu)_B^C | p_J \rangle_C \langle p_K |^D | r_N \rangle_D \langle r^N |^E (\Gamma_\mu)_E^F | p_L \rangle_F}{16(p \cdot r)^2} \\ &= \frac{\langle p_I | r_M \rangle \langle p_K | r_N \rangle}{16(p \cdot r)^2} \times \langle r^M |^B (\Gamma^\mu)_B^C | p_J \rangle_C \langle r^N |^E (\Gamma_\mu)_E^F | p_L \rangle_F \end{aligned} \quad (10)$$

After plugging the equation (8) into the second part of (10), it becomes

$$\begin{aligned} &\langle r^M |^B | p_J \rangle_C \langle r^N |^E | p_L \rangle_F (-2\Omega_{BE}\Omega^{CF} + 2\delta_B^F \delta_E^C - \delta_B^C \delta_E^F) \\ &= -2\langle r^M | r^N \rangle \langle p_L | p_J \rangle + 2\langle r^M | p_L \rangle \langle r^N | p_J \rangle - \langle r^M | p_J \rangle \langle r^N | p_L \rangle \end{aligned} \quad (11)$$

The first term of (11) equals to 0 because the equality

$$\langle p_I | p_J \rangle = 0 \quad (12)$$

and the remaining two term can be contracted with the first part of (10), which helps to obtain the following two terms

$$2\langle p_I | r_M \rangle \langle r^M | p_L \rangle \langle p_K | r_N \rangle \langle r^N | p_J \rangle - \langle p_I | r_M \rangle \langle r^M | p_J \rangle \langle p_K | r_N \rangle \langle r^N | p_L \rangle \quad (13)$$

It is known that

$$\langle p_I | (r \cdot \Gamma) | p_J \rangle = \text{Tr}[|p_J\rangle \langle p_I| (r \cdot \Gamma)] = \frac{1}{2}\epsilon_{IJ}\text{Tr}[(p \cdot \Gamma)(r \cdot \Gamma)] = 2\epsilon_{IJ}(p \cdot r) \quad (14)$$

so the formal equation can be rewritten like

$$8(p \cdot r)^2 \epsilon_{IL}\epsilon_{KJ} - 4(p \cdot r)^2 \epsilon_{IJ}\epsilon_{KL} \quad (15)$$

Finally, we need to symmetrize the indices (IJ) and (KL) , so that we obtain

$$\varepsilon_{IJ\varepsilon_{KL,\mu}}^\mu = \frac{1}{8(p \cdot r)^2} (p \cdot r)^2 (\epsilon_{IL}\epsilon_{KJ} + \epsilon_{JL}\epsilon_{KI} + \epsilon_{IK}\epsilon_{LJ} + \epsilon_{JK}\epsilon_{LI}) - \frac{1}{16(p \cdot r)^2} (p \cdot r)^2 (\epsilon_{(IJ)}\epsilon_{(KL)}) \quad (16)$$

because ϵ itself is an antisymmetric tensor so it will become zero under the symmetrization, so that the second term is vanishing.

After some rearrangement of indices, we can obtain

$$\varepsilon_{IJ}^\mu \varepsilon_{KL,\mu} = \frac{1}{4}(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{IK}\epsilon_{LJ}) \quad (17)$$

Unfortunately, when we want to reproduce the original normalization relation, we will obtain something like

$$\begin{aligned} \varepsilon_a^\mu \varepsilon_{b,\mu} &= (-\varepsilon_{(IJ)}^\mu (\sigma_a)^{IJ})(-\varepsilon_{(KL)}^\mu (\sigma_b)^{KL}) \\ &= \frac{1}{4}(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{IK}\epsilon_{LJ})(\sigma_a)^{IJ}(\sigma_b)^{KL} \\ &= \frac{1}{4}\epsilon_{IL}\epsilon_{KJ}\epsilon^{JM}(\sigma_a)^I{}_M\epsilon^{LN}(\sigma_b)^K{}_N + \frac{1}{4}\epsilon_{IK}\epsilon_{LJ}\epsilon^{IM}(\sigma_a)^J{}_M\epsilon^{LN}(\sigma_b)^K{}_N \\ &= \frac{1}{4}\delta_I^N\delta_K^M(\sigma_a)^I{}_M(\sigma_b)^K{}_N + \frac{1}{4}\delta_K^M\delta_J^N(\sigma_a)^J{}_M(\sigma_b)^K{}_N \\ &= \frac{1}{4}2\text{tr}(\sigma_a\sigma_b) \\ &= \delta_{ab} \end{aligned} \quad (18)$$

2 Completeness Relation of Polarization Vectors

For this purpose, we need to compute the following formula

$$\varepsilon_a^\mu \varepsilon^{a,\nu} = ? \quad (19)$$

If we multiply the momentum p_μ , it should become vanishing so that the completeness relation should look like

$$\varepsilon_a^\mu \varepsilon^{a,\nu} = k(-\eta^{\mu\nu} + \frac{p^\mu r^\nu + p^\nu r^\mu}{(p \cdot r)^2}) \quad (20)$$

here k is just a overall coefficient.

Here, we choose to compute the completeness relation for polarization vectors with fundamental indices

$$\varepsilon_{IJ}^\mu \varepsilon^{IJ,\nu} = a(-\eta^{\mu\nu} + \frac{p^\mu r^\nu + p^\nu r^\mu}{(p \cdot r)^2}) \quad (21)$$

with a different coefficient.

For symplisity, we start with the convention from paper arxiv:2202.08257, in which the polarization vector looks like

$$\epsilon_{IJ}^\mu = \frac{\langle p_I | \Gamma^\mu | r_J \rangle}{\sqrt{2}} \quad (22)$$

with the normalization

$$\langle p^I | r^J \rangle = \epsilon^{IJ} \quad (23)$$

which means the following condition

$$2(p \cdot r) = 1 \quad (24)$$

The method to compute this completeness relation is still just multiplying them together like

$$\varepsilon_{(IJ)}^\mu \varepsilon^{(IJ),\nu} = \frac{\langle p_I | \Gamma^\mu | r_J \rangle}{\sqrt{2}} \times \frac{\langle p^I | \Gamma^\nu | r^J \rangle}{\sqrt{2}} \quad (25)$$

It can be decomposed into 4 pices IJ^{IJ} , IJ^{JI} , JJ^{IJ} , JJ^{JI} , and the first one can be computed like

$$\begin{aligned} \langle p_I | \Gamma^\mu | r_J \rangle \times \langle p^I | \Gamma^\nu | r^J \rangle &= -\langle p_I | \Gamma^\mu | r_J \rangle \langle r^J | \Gamma^\nu | p^I \rangle \\ &= -\text{Tr}[[p^I] \langle p_I | \Gamma^\mu | r_J \rangle \langle r^J | \Gamma^\nu] \\ &= \text{Tr}[(p \cdot \Gamma) \Gamma^\mu (r \cdot \Gamma) \Gamma^\nu] \\ &= 4(p^\mu r^\nu + p^\nu r^\mu - (p \cdot r) \eta^{\mu\nu}) \end{aligned} \quad (26)$$

where we use a nontrivial fact that

$$\langle p^I | \Gamma^\nu | r^J \rangle = -\langle r^J | \Gamma^\nu | p^I \rangle \quad (\text{obtained from } C^T \Gamma^\mu C = -(\Gamma^\mu)^T) \quad (27)$$

The fourth piece is nearly the same as the first one

$$\langle p_J | \Gamma^\mu | r_I \rangle \times \langle p^J | \Gamma^\nu | r^I \rangle = 4(p^\mu r^\nu + p^\nu r^\mu - (p \cdot r) \eta^{\mu\nu}) \quad (28)$$

The remaining two need some extra efforts, here I compute it explicitly

$$\langle p_I | \Gamma^\mu | r_J \rangle \times \langle p^J | \Gamma^\nu | r^I \rangle \quad (29)$$

In terms of two massless spinors that satisfy (23), the $USp(2, 2)$ identity operator can be written as

$$|r_a\rangle_A \langle p^a|^B + |p_a\rangle_A \langle r^a|^B = \delta_A^B \quad (30)$$

It can be used to rewrite the equation (29) like

$$\begin{aligned} &\langle p_I |^A (\Gamma^\mu)_A^B |r_J\rangle_B \langle p^J |^C (\Gamma^\nu)_C^D |r^I\rangle_D \\ &= \langle p_I |^A (\Gamma^\mu)_A^B (\delta_B^C - |p_J\rangle_B \langle r^J|^C) (\Gamma^\nu)_C^D |r^I\rangle_D \\ &= \langle p_I |^A (\Gamma^\mu)_A^B (\Gamma^\nu)_B^D |r^I\rangle_D - \langle p_I | \Gamma^\mu | p_J \rangle \langle r^J | \Gamma^\nu | r^I \rangle \end{aligned} \quad (31)$$

And similarly for

$$\begin{aligned} \langle p_J | \Gamma^\mu | r_I \rangle \times \langle p^I | \Gamma^\nu | r^J \rangle &= \langle p^I | \Gamma^\nu | r^J \rangle \times \langle p_J | \Gamma^\mu | r_I \rangle \\ &= \langle p^I |^A (\Gamma^\nu)_A^B (-|p^J\rangle_B \langle r_J|^C - \delta_B^C) (\Gamma^\mu)_C^D |r_I\rangle_D \\ &= -\langle p^I |^A (\Gamma^\nu)_A^B (\Gamma^\mu)_B^D |r_I\rangle_D - \langle p^I | \Gamma^\nu | p^J \rangle \langle r_J | \Gamma^\mu | r_I \rangle \end{aligned} \quad (32)$$

Now we can combine these two terms

$$\begin{aligned} &\langle p_I |^A (\Gamma^\mu)_A^B (\Gamma^\nu)_B^D |r^I\rangle_D - \langle p_I | \Gamma^\mu | p_J \rangle \langle r^J | \Gamma^\nu | r^I \rangle - \langle p^I |^A (\Gamma^\nu)_A^B (\Gamma^\mu)_B^D |r_I\rangle_D - \langle p^I | \Gamma^\nu | p^J \rangle \langle r_J | \Gamma^\mu | r_I \rangle \\ &= \langle p_I | (\Gamma^\mu \Gamma^\nu + \Gamma^\nu \Gamma^\mu) | r^I \rangle - \langle p_I | \Gamma^\mu | p_J \rangle \langle r^J | \Gamma^\nu | r^I \rangle - \langle p^I | \Gamma^\nu | p^J \rangle \langle r_J | \Gamma^\mu | r_I \rangle \\ &= 2\eta^{\mu\nu} \langle p_I | r^I \rangle - (2p^\mu \epsilon_{IJ})(-2r^\nu \epsilon^{JI}) - (-2p^\nu \epsilon^{IJ})(2r^\mu \epsilon_{JI}) \\ &= 4\eta^{\mu\nu} + 8p^\mu r^\nu + 8p^\nu r^\mu \end{aligned} \quad (33)$$

Above all, we can conclude that

$$\begin{aligned} \varepsilon_{(IJ)}^\mu \varepsilon^{(IJ),\nu} &= \frac{1}{2} \times \frac{1}{8} [8(p^\mu r^\nu + p^\nu r^\mu - (p \cdot r) \eta^{\mu\nu}) + 4\eta^{\mu\nu} + 8p^\mu r^\nu + 8p^\nu r^\mu] \\ &\stackrel{?}{=} 2p^{(\mu} r^{\nu)} \end{aligned} \quad (34)$$

3 Completeness Relation Review

Under the following definition of polarization vector

$$\varepsilon_a^\mu = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_J \rangle}{4(p \cdot r)} (\sigma_a)_K^I \epsilon^{KJ} \quad (35)$$

and same for

$$\varepsilon_a^\mu = \frac{\langle p^I | (r \cdot \Gamma) \Gamma^\mu | p^J \rangle}{4(p \cdot r)} (\sigma_a)_J^K \epsilon_{KI}, \quad (36)$$

the polarization vector living in the adjoint representation can be defined as

$$\varepsilon_{IJ}^\mu(p; r) := \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_J \rangle}{4(p \cdot r)} \quad (37)$$

$$\varepsilon^{IJ,\mu}(p; r) := \frac{\langle p^I | (r \cdot \Gamma) \Gamma^\mu | p^J \rangle}{4(p \cdot r)}. \quad (38)$$

The corresponding complex conjugate can be obtained

$$(\varepsilon_{IJ}^\mu(p; r))^* = -\frac{\langle p^I | \Gamma^\mu(r \cdot \Gamma) | p^J \rangle}{4(p \cdot r)} \quad (39)$$

and similarly

$$(\varepsilon^{IJ,\mu}(p; r))^* = -\frac{\langle p_I | \Gamma^\mu(r \cdot \Gamma) | p_J \rangle}{4(p \cdot r)}. \quad (40)$$

For the consistency of index placement with the summation convention, we can define the following quantities

$$\varepsilon_{IJ}^{*\mu}(p; r) := (\varepsilon^{IJ,\mu}(p; r))^* = -\frac{\langle p_I | \Gamma^\mu(r \cdot \Gamma) | p_J \rangle}{4(p \cdot r)}, \quad (41)$$

$$\varepsilon^{*IJ,\mu}(p; r) := (\varepsilon_{IJ}^\mu(p; r))^* = -\frac{\langle p^I | \Gamma^\mu(r \cdot \Gamma) | p^J \rangle}{4(p \cdot r)} \quad (42)$$

Then the completeness relation can be computed like

$$\varepsilon^{*IJ,\mu} \varepsilon_{IJ}^\nu = -\frac{\langle p^I | \Gamma^\mu(r \cdot \Gamma) | p^J \rangle}{4(p \cdot r)} \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\nu | p_J \rangle}{4(p \cdot r)} \quad (43)$$

It includes 4 terms and can be grouped into two part, the first part can be computed like

$$\begin{aligned} & \langle p^I | \Gamma^\mu(r \cdot \Gamma) | p^J \rangle \langle p_I | (r \cdot \Gamma) \Gamma^\nu | p_J \rangle \\ &= -\langle p^I | \Gamma^\mu(r \cdot \Gamma) | p^J \rangle \langle p_J | \Gamma^\nu(r \cdot \Gamma) | p_I \rangle \\ &= -\text{tr}[\langle p_I | \langle p^I | \Gamma^\mu(r \cdot \Gamma) | p^J \rangle \langle p_J | \Gamma^\nu(r \cdot \Gamma)] \\ &= \text{tr}[(p \cdot \Gamma) \Gamma^\mu(r \cdot \Gamma) (p \cdot \Gamma) \Gamma^\nu(r \cdot \Gamma)] \end{aligned} \quad (44)$$

Here we need to compute trace for the product of 6 gamma matrices, and I just give the answer

$$\begin{aligned} \text{Tr}(\Gamma^a \Gamma^b \Gamma^c \Gamma^d \Gamma^e \Gamma^f) = 4 \big(& \eta^{ab} \eta^{cd} \eta^{ef} - \eta^{ab} \eta^{ce} \eta^{df} + \eta^{ab} \eta^{cf} \eta^{de} \\ & - \eta^{ac} \eta^{bd} \eta^{ef} + \eta^{ac} \eta^{be} \eta^{df} - \eta^{ac} \eta^{bf} \eta^{de} \\ & + \eta^{ad} \eta^{bc} \eta^{ef} - \eta^{ad} \eta^{be} \eta^{cf} + \eta^{ad} \eta^{bf} \eta^{ce} \\ & - \eta^{ae} \eta^{bc} \eta^{df} + \eta^{ae} \eta^{bd} \eta^{cf} - \eta^{ae} \eta^{bf} \eta^{cd} \\ & + \eta^{af} \eta^{bc} \eta^{de} - \eta^{af} \eta^{bd} \eta^{ce} + \eta^{af} \eta^{be} \eta^{cd} \big). \end{aligned} \quad (45)$$

A compact way to write the result is

$$\text{Tr}(\Gamma^{a_1} \dots \Gamma^{a_6}) = 4 \sum_{\text{pairings}} (-1)^\sigma \eta^{a_{i_1} a_{j_1}} \eta^{a_{i_2} a_{j_2}} \eta^{a_{i_3} a_{j_3}}.$$

Here the sum runs over all pairwise partitions (pairings) of the index set $\{1, \dots, 6\}$ into three disjoint pairs $\{(i_1, j_1), (i_2, j_2), (i_3, j_3)\}$. Here $(-1)^\sigma$ denotes the sign (parity) of the permutation needed to reorder the sequence so that the members of each pair become adjacent.

Using this formula, we can rewrite the equation (44)

$$\begin{aligned} (44) = & p^\mu r^\nu (p \cdot r) - p^\mu r^\nu (p \cdot r) + p^\mu p^\nu r^2 \\ & - p^\mu r^\nu (p \cdot r) + \eta^{\mu\nu} (p \cdot r)^2 - p^\nu r^\mu (p \cdot r) \\ & + r^\mu r^\nu p^2 - \eta^{\mu\nu} p^2 r^2 + r^\mu r^\nu p^2 \\ & - p^\nu r^\mu (p \cdot r) + p^\nu p^\mu r^2 - p^\nu r^\mu (p \cdot r) \\ & + p^\nu r^\mu (p \cdot r) - p^\mu r^\nu (p \cdot r) + \eta^{\mu\nu} (p \cdot r)^2 \\ & = 4(-2p^\mu r^\nu (p \cdot r) - 2p^\nu r^\mu (p \cdot r) + \eta^{\mu\nu} (p \cdot r)^2) \end{aligned} \quad (46)$$

in which many terms cancel with each other.

Similarly, the second part can be computed like

$$\begin{aligned} & \langle p^I | \Gamma^\mu (r \cdot \Gamma) | p^J \rangle \langle p_J | (r \cdot \Gamma) \Gamma^\nu | p_I \rangle \\ = & \text{tr} [| p_I \rangle \langle p^I | \Gamma^\mu (r \cdot \Gamma) | p^J \rangle \langle p_J | (r \cdot \Gamma) \Gamma^\nu] \\ = & -\text{tr} [(p \cdot \Gamma) \Gamma^\mu (r \cdot \Gamma) (p \cdot \Gamma) (r \cdot \Gamma) \Gamma^\nu] \\ = & -\text{tr} [(p \cdot \Gamma) \Gamma^\mu (r \cdot \Gamma) (p \cdot \Gamma) \{ (r \cdot \Gamma), \Gamma^\nu \}] + \text{tr} [(p \cdot \Gamma) \Gamma^\mu (r \cdot \Gamma) (p \cdot \Gamma) \Gamma^\nu (r \cdot \Gamma)] \end{aligned} \quad (47)$$

It can be noticed that the first term equals to 0 and the second term is just the same as the result for first part. In short, the two part give us the same result.

Above all, the completeness relation can be written as

$$\begin{aligned} \varepsilon^{*IJ,\mu} \varepsilon_{IJ}^\nu &= -\frac{1}{16(p \cdot r)^2} \times \frac{1}{4} \times [4 \times 4(-2p^\mu r^\nu (p \cdot r) - 2p^\nu r^\mu (p \cdot r) + \eta^{\mu\nu} (p \cdot r)^2)] \\ &= -\frac{1}{2} \left[\frac{-p^\mu r^\nu - p^\nu r^\mu}{(p \cdot r)} + \eta^{\mu\nu} \right] \\ &= -\frac{1}{2} \eta^{\mu\nu} + \frac{p^{(\mu} r^{\nu)}}{(p \cdot r)} \end{aligned} \quad (48)$$

We can quickly check its correctness by multiplying a momentum

$$p_\mu \varepsilon^{*IJ,\mu} \varepsilon_{IJ}^\nu = -\frac{1}{2} p^\nu + \frac{\frac{1}{2} p^2 r^\nu + \frac{1}{2} p^\nu (p \cdot r)}{(p \cdot r)} = 0 \quad (49)$$